

BMU Test Plan — How to Read This Workbook

Purpose	Walk-through-ready test case list for the BMU verification project. Each row is one test case:
---------	--

Column	Meaning
Test ID	Unique identifier, grouped by operation/category prefix (e.g. TC_OR_001).
Feature / Operation	The BMU Specification v1.2 section this test targets.
Category	Directed-Valid / Directed-Corner / Directed-Error / Timing / Reset / CSR / Random / Scope-
Priority	High = blocks sign-off if failing. Medium = important, not blocking. Low = nice-to-have /
Description	What functionality is being exercised and why this case matters — written so a reviewer
Stimulus Summary	The control-field and operand values applied (abbreviated; full values live in the sequence
Expected Result	result_ff / error expected, per BMU Specification v1.2 (or explicitly marked
Coverage Bin	The functional coverage bin this test is expected to close, per the Verification Plan
Sequence / File	Planned tb/sequences/ file implementing this test (naming matches the repo structure).
Status	Not Started / In Progress / Passing / Failing / Blocked. Keep this current — the Summary
Notes	Anything a reader needs that doesn't fit elsewhere — bug IDs, clarification-log

<u>Category meanings</u>	
Directed-Valid	Nominal, spec-correct stimulus — confirms the operation produces the right result under
Directed-Corner	Boundary/edge operand values (0, all-1s, sign boundary, max shift amount, etc.) chosen
Directed-Error	Deliberate guard-condition violation — confirms result is forced to 0 and error=1 as
Timing	Cycle-accurate latency / valid_in-gating behavior (Specification §3).
Reset	Reset behavior, including mid-operation reset.
CSR	CSR bypass-read and CSR-write behavior, and the CSR-conflict chip-wide error condition.
Random	Constrained-random regression — broad confidence beyond the directed list.
Scope-TBD	Control-struct fields with no behavior specification (Specification §8). Not testable

Related documents	BMU_Specification_v1.2.docx (behavior source of truth) · BMU_Verification_Plan.docx
--------------------------	---